

NONCLINICAL REPORT

Nonclinical Report No. MOD-3938 and MOD-3940

Evaluation of Immunogenicity and Determination of Titers Dynamic Range of mRNA-1273
(SARS-CoV-2)

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LIST OF ABBREVIATIONS

Abbreviation	Definition
ASC	antibody-secreting cell
BD	Becton, Dickinson and Company
CoV	coronavirus
DSPC	1,2-distearoyl-sn-glycero-3-phosphocholine
ELISA	enzyme-linked immunosorbent assay
ELISpot	enzyme-linked immunospot
FBS	fetal bovine serum
FDA	Food and Drug Administration
GLP	Good Laboratory Practice
GMP	Good Manufacturing Practice
HEPES	4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid
HI	heat inactivated
IACUC	Institutional Animal Care and Use Committee
IC ₅₀	half-maximal inhibitory concentration
IFN	interferon
IgG	immunoglobulin G
IL	interleukin
IM	intramuscular
LN	lymph node
LNP	lipid nanoparticle
mRNA	messenger RNA
NTD	N-terminal domain
OD	optical density
PBS	phosphate-buffered saline
PEG	polyethylene glycol
PEG-2000DMG	1-monomethoxypolyethyleneglycol-2,3-dimyristylglycerol polyethylene glycol
RBD	receptor binding domain
S	spike
S-2P	spike protein with 2 proline substitutions within the heptad repeat 1 domain
SARS-CoV-2	2019 novel coronavirus
SM-102	heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6(undecyloxy)hexyl)amino)octanoate
Th1	T-helper 1
Th2	T-helper 2
TNF- α	tumor necrosis factor alpha
UTR	untranslated region

QUALITY ASSURANCE STATEMENT

The studies were not conducted according to Good Laboratory Practice (GLP) regulations.

1. SUMMARY

Nonclinical studies have demonstrated that coronavirus (CoV) spike (S) proteins are immunogenic and that 2019 novel coronavirus (SARS-CoV-2) prefusion-stabilized S protein with 2 proline substitutions within the heptad repeat 1 domain (S-2P)-based vaccines, including those based on messenger RNA (mRNA) delivery platforms, are protective in animals. In addition, previous clinical studies of vaccines targeting related CoVs and other viruses have demonstrated that mRNA-based vaccines are safe and immunogenic.

Humoral and cellular immune responses were evaluated from a wide range of mRNA-1273 doses administered via intramuscular (IM) injection on a prime/boost schedule in young BALB/c mice in 2 studies (Study MOD-3938 and Study MOD-3940). The objectives of these studies were to evaluate the immunogenicity and determine the titer dynamic range of mRNA-1273 (Study MOD-3938), and repeat for confirmation of dose responses (Study MOD-3940). The 2 independent non-GLP mouse studies were conducted with the aims of assessing an immunological endpoint in a robust and reproducible way with statistical power for comparison and understanding both dose response and inter- and intra-study variability.

ModernaTX, Inc. (Sponsor) is developing mRNA-1273, a novel lipid nanoparticle (LNP)-encapsulated mRNA-based vaccine against SARS-CoV-2. mRNA-1273 contains a single mRNA that encodes for the SARS-CoV-2 S glycoprotein stabilized in the prefusion conformation with 2 proline substitutions within the heptad repeat 1 domain (S-2P). The mRNA-1273 Drug Product consists of the mRNA formulated in a mixture of 4 ionizable, structural, helper, and polyethylene glycol (PEG) lipids common to the Sponsor's, mRNA vaccine platform: heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy) hexyl)amino)octanoate (SM-102); cholesterol; 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC); and 1-monomethoxypolyethyleneglycol-2,3-dimyristylglycerol polyethylene glycol (PEG2000-DMG), respectively. The test article in this study consisted of mRNA-1273 Drug Product (WH2020_NatSP_2P, Tech dev DP) and phosphate-buffered saline (PBS) as a negative control.

Groups of BALB/c mice were vaccinated with multiple dose levels of mRNA-1273 (20, 10, 5, 2.5, 1.25, 0.63, 0.32, 0.16, 0.08, 0.04, 0.02, 0.01, 0.005, or 0.0025 µg) or PBS, administered via IM injection on a prime (Week 0)/boost (Week 3) schedule. In both studies, sera were collected at Week 3 (Day 21; before the second injection) and Week 5 (Day 36) for assessment of SARS-CoV-2 S protein-specific immunoglobulin G (IgG) by enzyme-linked immunosorbent assay (ELISA).

In addition to analysis of S binding titers, identification of antibodies that bind to the immunodominant subdomains of the S protein, specifically the receptor binding domain (RBD), the N-terminal domain (NTD) and the S1 subunit, can demonstrate response against these important functional components of the S protein. In Study MOD-3938, 2 weeks post-boost, sera

were analyzed for RBD, NTD, and S1 subunit-specific IgG by ELISA. As it is important to determine the functional properties of the antibodies elicited by vaccination, neutralizing activity against a homotypic SARS-CoV-2 pseudovirus was also evaluated in sera 2 weeks post-boost.

In Study MOD-3940 only, spleens and draining lymph nodes (LNs) harvested on Day 36 were analyzed for antigen-reactive IgG antibody-secreting cells (ASCs) by enzyme-linked immunospot (ELISpot) assay using the entire S protein as well as the S1 and S2 subunit proteins.

In Study MOD-3938, spleens were collected at 2 weeks post-boost from mice in the 10 highest dose levels of mRNA-1273 and from mice in the PBS group. Splenocytes were stimulated using 2 S peptide pools that, in combination, cover the entire SARS-CoV-2 S protein and T-cell analyses were performed by intracellular staining and flow cytometry.

BALB/c mice immunized with varying levels of mRNA-1273 exhibited consistency across the 2 independent studies, with similar half-maximal inhibitory concentration (IC_{50}) and SARS-CoV-2 S protein-specific IgG binding antibody titer levels. The range of titers within a group was smaller at higher dose levels and larger at lower dose levels.

In Study MOD-3938, mRNA-1273 elicited a dose-dependent neutralizing antibody response and there was a strong correlation between binding and neutralizing antibody responses.

Also in Study MOD-3938, a dose-dependent response in binding IgG antibodies against the RBD, NTD, and S1 subunits of the SARS-CoV-2 S protein was observed. Binding IgG antibodies against S1 and RBD were detected even at the low dose of 0.01 μ g of mRNA-1273.

In Study MOD-3940, mRNA-1273 induced SARS-CoV-2 whole S protein and S1 and S2 subunit-specific IgG antigen-reactive ASCs both in the spleen and LNs in a dose-dependent manner. Responses to the S2 subunit were lower than those against the S1 subunit of the SARS-CoV-2 S protein.

In Study MOD-3938, immunization with mRNA-1273 produced S protein-specific CD4⁺ and CD8⁺ T-cell responses in splenocytes stimulated with S1 and S2 subunit overlapping peptide pools encompassing the entire SARS-CoV-2 S protein. The CD4⁺ T-cell responses had a T-helper 1 (Th1)-directed cytokine profile as indicated by elevated levels of interferon [IFN]- γ , interleukin (IL)-2, and tumor necrosis factor alpha (TNF- α). Low CD4⁺ Th2 cytokine levels were detected in mice immunized with mRNA-1273 across all doses. A dose-dependent CD4⁺ Th1 cytokine response was not observed. Spike antigen-specific CD8⁺ responses decreased with decreasing dose levels.

In conclusion, mRNA-1273 is immunogenic in BALB/c mice, showing a dose-dependent response in SARS-CoV-2 whole S protein-specific IgG binding antibody titers that was similar across the 2 studies (MOD-3938 and MOD-3940). In addition, RBD, NTD, and S1 subunit-specific IgG

antibody titers as well as neutralization titers measured in Study MOD-3938 showed similar dose-dependent responses. There was a strong correlation between binding and neutralizing antibody responses. In Study MOD-3940, mRNA-1273 induced IgG antigen-reactive ASCs in both the spleen and LNs in a dose-dependent manner. In Study MOD-3938, in splenocytes stimulated with S1 and S2 subunit overlapping peptide pools encompassing the entire SARS-CoV-2 S protein, mRNA-1273 drove Th1-directed CD4⁺ T-cell responses and CD8⁺ T-cell responses. Although a dose-dependent CD4⁺ Th1 cytokine response was not observed, CD8⁺ T-cell response decreased with decreasing dose.

2. DESCRIPTION OF TEST PROCEDURES

2.1. Conduct of Study

2.1.1. Sponsor

ModernaTX, Inc.
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2.1.2. Performing Laboratory

ModernaTX, Inc.
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Cambridge, MA 02139

2.1.3. Study Numbers

Study numbers MOD-3938 and MOD-3940.

2.1.4. Purpose of the Study

The objectives of these studies were to evaluate the immunogenicity and determine the titer dynamic range of mRNA-1273 (Study MOD-3938), repeated for confirmation of dose responses (Study MOD-3940), in young BALB/c mice.

2.1.5. Institutional Animal Care and Use Committee Approval

All animal experiments described in this study report were approved and conducted in accordance with the regulations of the Sponsor. Institutional Animal Care and Use Committee (IACUC Protocol 18-03-04).

2.1.6. Study Director

(b) (6)

ModernaTX, Inc.
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Cambridge, MA 02139

2.1.7. Key Personnel

Role	Study MOD-3938	Study MOD-3940
Study coordinator	(b) (6) ModernaTX, Inc. (b) (6) Diseases ModernaTX, Inc.	(b) (6) ModernaTX, Inc. (b) (6) Diseases ModernaTX, Inc.
Study execution group leader	(b) (6) ModernaTX, Inc.	(b) (6) ModernaTX, Inc.
Analytical assays, ELISA lead	(b) (6) Infectious Disease, ModernaTX, Inc.	(b) (6) Infectious Disease, ModernaTX, Inc.
Analytical assays, T-cell assay lead	(b) (6) Assay Development, Infectious Disease ModernaTX, Inc.	(b) (6) Assay Development, Infectious Disease ModernaTX, Inc.
Analytical assays, ELISpot lead	(b) (6) Infectious Disease ModernaTX, Inc.	(b) (6) Infectious Disease ModernaTX, Inc.

2.1.8. Dates of Study Performance

Schedule of Events	Study MOD-3938	Study MOD-3940
Test administration Prime dose Boost dose	09 Apr 2020 (Study Day 1) 30 Apr 2020 (Study Day 22)	22 Apr 2020 (Study Day 1) 13 May 2020 (Study Day 22)
Serum collection	29 Apr 2020 (Study Day 21) 14 May 2020 (Study Day 36)	12 May 2020 (Study Day 21) 27 May 2020 (Study Day 36)
Spleen and lymph node collection	14 May 2020 (Study Day 36)	27 May 2020 (Study Day 36)

2.1.9. Records Maintained

Data are archived in the Sponsor Electronic Notebook System at the following location:

Study MOD-3938 Electronic Notebook Project: Moderna Informatics Digital In Vivo Studies Application

Study MOD-3938 Electronic Notebook Experiment: Moderna Informatics Digital In Vivo Studies Application

Study MOD-3940 Electronic Notebook Project: Moderna Informatics Digital In Vivo Studies Application

Study MOD-3940 Electronic Notebook Experiment: Moderna Informatics Digital In Vivo Studies Application

2.2. Test Articles

The test article included multiple dose levels of mRNA-1273. Phosphate-buffered saline was used as a negative control.

2.2.1. mRNA-1273 Test Article

mRNA-1273 contains a single mRNA that encodes SARS-CoV-2 S-2P formulated into a mixture of 4 lipids: SM-102; cholesterol; DSPC; and PEG2000-DMG. A summary of the dose formulation analysis results is presented in [Table 1](#).

The mRNA-1273 Drug Product used in these studies was an mRNA formulation (WH2020_NatSP_2P, Tech dev DP, AMPDP-200005) prepared with a same method as the Good Manufacturing Practice (GMP) mRNA-1273 clinical Drug Product ([Appendix 1](#)). This product is a novel LNP-encapsulated mRNA-based vaccine against SARS-CoV-2. mRNA-1273 encodes for the full-length S protein of SARS-CoV-2, modified to introduce 2 proline residues to stabilize the S protein into a prefusion conformation (S-2P). mRNA-1273 consists of mRNA encapsulated in an LNP through a modified ethanol-drop nanoprecipitation process described previously ([Hassett et al 2019](#)).

Preclinical mRNA-1273 vaccine was manufactured by the Sponsor. A sequence-optimized mRNA encoding prefusion-stabilized SARS-CoV-2 S-2P protein was synthesized in vitro using an optimized T7 RNA polymerase-mediated transcription reaction with complete replacement of uridine by N1m-pseudouridine ([Nelson et al 2020](#)). The reaction included a DNA template containing the immunogen open-reading frame flanked by 5' untranslated region (UTR) and 3' UTR sequences and was terminated by an encoded polyA tail. After transcription, the Cap 1 structure was added to the 5' end using vaccinia capping enzyme (New England Biolabs) and vaccinia 2'O-methyltransferase (New England Biolabs). The mRNA was purified (b) (4)

(b) (4) buffer exchanged (b) (4) sterile filtered, and kept frozen at -20°C until further use.

mRNA-1273 was encapsulated in an LNP through a modified ethanol-drop nanoprecipitation process as previously described ([Hassett et al 2019](#)). (b) (4)

The mixture was neutralized with tris(hydroxymethyl)aminomethane-hydrochloride, pH 7.5, sucrose was added as a cryoprotectant, and the final solution was sterile filtered. Vials were filled with mRNA-1273 formulated LNP and stored frozen at -70°C until further use. The Drug Product underwent analytical characterization, which included the determination of particle size and polydispersity, encapsulation, mRNA purity, double-stranded RNA content, osmolality, pH, endotoxin content, and bioburden, and the material was deemed acceptable for the in vivo study ([Corbett et al 2020](#)).

Table 1: Test Materials and Summary of Dose Formulation Analysis Results

Test Material	mRNA Description	LNP Lot No.	mRNA Lot No.	Size (nm)	PDI	EE %
mRNA-1273 Drug Product	mRNA SARS-CoV-2 spike protein ^a	(b) (4)				

Abbreviations: EE% = encapsulation efficiency ; LNP = lipid nanoparticle; mRNA = messenger RNA; PDI = polydispersity index; SARS-CoV-2 = 2019 novel coronavirus.

Note: All mRNA test material stocks were formulated in SM-102-containing lipid nanoparticles with a final storage buffer in 10.7 mM sodium acetate; 8.7% sucrose; 20 mM tris(hydroxymethyl)aminomethane hydrochloride at pH 7.5 and stored frozen at -70°C until further use.

The EE% is the percentage of mRNA (b) (4)

(b) (4)

^a Full-length SARS-CoV-2 spike protein stabilized in the prefusion conformation with 2 proline substitutions within the heptad repeat 1 domain (S-2P) and including a transmembrane domain for insertion into the cell membrane after translation.

^b Manufactured by ModernaTX, Inc. (01 Feb 2020).

2.2.2. Phosphate-Buffered Saline

Phosphate-buffered saline (b) (4)) was purchased commercially.

2.3. Test System

2.3.1. Mice

Female BALB/c mice, approximately 8 to 9 weeks old, were obtained from Charles River Laboratories International, Inc. (Kingston, NY).

2.4. Methods

2.4.1. Study Design and Treatment Groups

A key goal of the mRNA in vivo workstream was to define the animal study design and immunological endpoint for batch comparison and in-vitro-to-in-vivo concordance studies. Parameters to define included mouse strain, dose levels, number of animals per group, time point post-vaccine administration, and type/format of serological assay. The set of chosen conditions was selected in order to yield robust and reproducible data with statistical power for comparison. The immunogenicity and determination of titer dynamic range of mRNA-1273 were evaluated in BALB/c mice (Study MOD-3938) and repeated for variability testing (Study MOD-3940). BALB/c mice were selected for mRNA-1273 based on past data.

Mice were treated according to the study design presented in [Table 2](#) and [Table 3](#) for Studies MOD-3938 and MOD-3940, respectively. In both studies, groups of female BALB/c mice were administered mRNA-1273 (20, 10, 5, 2.5, 1.25, 0.63, 0.32, 0.16, 0.08, 0.04, 0.02, 0.01, 0.005, or 0.0025 µg) or PBS. Mice were vaccinated via IM injection in the right hind leg on a prime (Week 0)/boost (Week 3) administration schedule. Sera were collected at Week 3 (Day 21; before the second injection) and Week 5 (Day 36) from all animals and measured for SARS-CoV-2 S protein-specific IgG by ELISA.

In Study MOD-3938 only ([Table 2](#)), Week 5 (Day 36; 2 weeks post-boost) sera were analyzed for RBD, NTD, and S1 subunit-specific IgG by ELISA. Week 5 sera (for the 20 through 0.02 µg groups only) was also measured for neutralizing antibody titers against a homotypic SARS-CoV-2 pseudovirus. Spleens were collected 2 weeks post-boost (Day 36) from mice in the 10 highest dose levels of mRNA-1273 (20, 10, 5, 2.5, 1.25, 0.63, 0.32, 0.16, 0.08 and 0.04 µg) and from mice in the PBS group.

In Study MOD-3940 only ([Table 3](#)), at 2 weeks post-boost, antigen-reactive IgG ASCs collected from spleens and draining LNs in animals dosed with 10 and 1.25 µg of mRNA-1273 and PBS were evaluated by ELISpot.

Table 2: Treatment Regimen for Study MOD-3938

Group	Number of Mice (N = 112)	Treatment	Dose (µg)	ROA (Volume)	Vaccination Schedule	Sample Collection
1	8	mRNA-1273	20	IM injection (50 µL)	<u>Prime:</u> Week 0/Day 1 <u>Boost:</u> Week 3/Day 22	Week 3/Day 21 and Week 5/Day 36 (sera for ELISA and neutralizing assay) Week 5/Day 36 (splens for cytokine analysis)
2	8		10			
3	8		5			
4	8		2.5			
5	8		1.25			
6	8		0.63			
7	8		0.32			
8	8		0.16			
9	4		0.08			
10	8		0.04			
11	8		0.02			
12	8		0.01			
13	8		0.005			
14	8		0.0025			
15	4	PBS	NA	IM injection (50 µL)		

Abbreviations: ELISA = enzyme-linked immunosorbent assay; IM = intramuscular; NA = not applicable; PBS = phosphate-buffered saline; ROA = route of administration.

Table 3: Treatment Regimen for Study MOD-3940

Group	Number of Mice (N = 120)	Treatment	Dose (µg)	ROA (Volume)	Vaccination Schedule	Sample Collection
1	8	mRNA-1273	20	IM injection (50 µL)	<u>Prime:</u> Week 0/Day 1 <u>Boost:</u> Week 3/Day 22	Week 3/Day 21 and Week 5/Day 36 (sera for ELISA) Week 5/Day 36 (lymph nodes and spleens for ELISpot)
2	8		10			
3	8		5			
4	8		2.5			
5	8		1.25			
6	8		0.63			
7	8		0.32			
8	8		0.16			
9	8		0.08			
10	8		0.04			
11	8		0.02			
12	8		0.01			
13	8		0.005			
14	8		0.0025			
15	8	PBS	NA	IM injection (50 µL)		

Abbreviations: ELISA = enzyme-linked immunosorbent assay; ELISpot = enzyme-linked immunospot; IM = intramuscular; NA = not applicable; PBS = phosphate-buffered saline; ROA = route of administration.

2.4.2. SARS-CoV-2 Spike, RBD Protein Subdomain, and NTD Protein Subdomain Enzyme-Linked Immunosorbent Assay

(b) (4) ELISA plates (b) (4) were coated with 100 µL of 1 µg/mL of SARS-CoV-2 S protein (S1+S2 Sino Biological #40589-V08B1) or 1 µg/mL of SARS-CoV-2 S1 subunit (Sino Biological #40592-V08H0), 1 µg/mL of SARS-CoV-2 RBD (Sino Biological #40592-V08H0), or 2 µg/mL of NTD subdomain protein prepared by the Sponsor. After overnight incubation at 4°C, plates were washed 4 times with PBS containing 0.05% Tween-20 and blocked for 2 hours at 37°C (b) (4). After washing, 5-fold serial dilutions of mouse serum were added (assay diluent: PBS + 0.05% Tween-20 + 5% goat serum). Plates were incubated for 2 hours at 37°C and washed, and horseradish peroxidase-conjugated goat anti-mouse IgG (b) (4) was added at a 1:10,000 dilution in assay diluent. Plates were incubated at 37°C for 1 hour, and then tetramethylbenzidine substrate (b) (4) was used to detect bound antibody. After incubation for 10 minutes at room temperature, the reaction was stopped by adding tetramethylbenzidine stop solution (b) (4) and the absorbance was measured at optical density (OD) 450 nm. Data were analyzed as indicated in [Section 2.5](#).

2.4.3. Lentivirus-Based Pseudovirus Neutralization Assay

(b) (4)

(b) (4)

and then pseudovirus neutralization assays were performed ([Pallesen et al 2017](#)). Briefly, heat-inactivated (HI) serum was mixed with pseudoviruses, incubated, and then added to angiotensin-converting enzyme 2 receptor-expressing HEK293T cells, obtained from (b) (6) at Scripps Research Institute. Cells were lysed after 72 hours of incubation, and luciferase activity (relative light units) was measured using a (b) (4) Luminometer by (b) (4) ([Corbett et al 2020](#)).

2.4.4. Enzyme-Linked Immunospot (ELISpot) Assay

Ninety-six-well ELISpot plates (b) (4) were coated overnight at 4°C with 100 µL of various recombinant proteins at 2 µg/ml (SARS-CoV-2 spike S1+ S2 protein, Sino Biological #40589-V08B1, SARS-CoV-2 S1 subunit, Sino Biological #40591-V08H, SARS-CoV-2 S2 subunit, Sino Biological #40590-V08B), all diluted in PBS. Mice were sacrificed at Day 14 post-boost, and the spleen and draining LNs were collected. Single-cell suspensions were prepared from fresh spleens in complete medium and used for ELISpot. The plates were washed in PBS and blocked with complete medium before a 2-fold serial dilution of cell suspension was added, starting at 0.5×10^6 for the spleens and 0.25×10^6 cells for the LNs. After incubation overnight at 37°C, 5% CO₂, the plates were washed in PBS with 0.05% Tween 20, and IgG-secreting cells were detected by biotinylated goat anti-mouse IgG (b) (4) (b) (4) followed by streptavidin-alkaline phosphatase (b) (4) and nitro blue tetrazolium/5-bromo-4-chloro-indolyl-phosphate (b) (4) substrate. The plates were imaged (b) (4) and spots manually counted.

2.4.5. T-Cell Stimulation and Intracellular Cytokine Staining

Whole mouse spleens were collected from female BALB/c mice on Day 36. A gentle (b) (4) tissue dissociator (b) (4) was used to generate mononuclear single-cell suspensions from whole mouse spleens. Following tissue dissociation, the cells were sieved through a 70 µm filter, and then density gradient centrifugation was performed using Fico/Lite-LM medium (b) (4) (b) (4). Cells from each mouse were resuspended in R10 media (b) (4) (b) (4) medium supplemented with Penicillin/Streptomycin antibiotic, 10% HI-fetal bovine serum [FBS], Glutamax, and N-2-hydroxyethylpiperazine-N-ethanesulfonic acid) and incubated at 37°C for 6 hours with protein transport inhibitor cocktail (b) (4) and 2 µg/mL of S peptide

pools (b) (4)). Control cells were incubated with no peptide stimulation. Cells were then washed with PBS prior to staining with (b) (4) Fixable Blue Dead Cell Stain (b) (4)) for 20 minutes at room temperature. Cells were subsequently washed with FC buffer (PBS supplemented with 2% HI-FBS and 0.05% sodium azide) and resuspended in (b) (4) (b) (4) FC Block (clone 2.4G2) for 5 minutes at room temperature. Staining was then performed with a surface stain cocktail containing the following antibodies: I-A/I-E PE (b) (4) CD8a BUV805 (b) (4) (b) (4) CD44 BUV395 (b) (4) CD62L BV605 (b) (4) (b) (4)), and CD4 BV480 (b) (4)) in brilliant stain buffer (b) (4)). Cells were subsequently washed with FC buffer then fixed using the (b) (4) solution kit and permeabilized using the (b) (4) solution kit. Kits were used according to the manufacturer's instructions. Cells were then washed with permeabilized and wash solution, and stained with FC Block for 5 minutes at room temperature prior to intracellular staining. Intracellular staining was then performed at 4°C for 30 minutes using a cocktail of antibodies: CD3e BUV737 (b) (4)), IFN- γ BV650 (b) (4)), (b) (4) TNF- α BV711 (b) (4)), IL-2 BV421 (b) (4)), IL-4 Alexa Fluor 488 (b) (4) (b) (4)), and IL-5 APC (b) (4)) in 1 x perm/wash diluted with brilliant stain buffer. Finally, cells were washed with permeabilized and wash solution and were resuspended in 0.5% paraformaldehyde-FC stain buffer prior to running on a (b) (4) (b) (4) flow cytometer (b) (4)) (Corbett et al 2020). Analysis was done using FlowJo software (version 10.6.2). Background cytokine expression in the no-peptide condition was subtracted from that measured in the peptide pools for each individual mouse.

2.5. Data Presentation and Analysis

Geometric means or means are represented by the heights of bars or symbols, and error bars represent the corresponding SD. Dotted lines indicate assay limits of detection. Mann-Whitney tests were used to compare 2 experimental groups, and Wilcoxon signed rank tests were used to compare the same animals at different time points. To compare >2 experimental groups, Kruskal-Wallis analysis of variance with Dunn's multiple comparisons tests were applied. For antibody responses, titers were determined using a 4-parameter logistic curve fit in GraphPad Prism (GraphPad Software, Inc.) and defined as the reciprocal dilution at approximately OD_{450nm} = 1.0 (normalized to a mouse standard on each plate). A Spearman's correlation was used to correlate binding antibody titers to neutralizing antibody titers (**p* value < .05, ***p* value < .01, ****p* value < .001, *****p* value < .0001). Percent neutralization was normalized considering uninfected cells as 100% neutralization and cells infected with only pseudovirus as 0%

neutralization. The IC_{50} titers were determined using a log (agonist) versus normalized response (variable slope) nonlinear function in Prism v8 (GraphPad) ([Corbett et al 2020](#)).

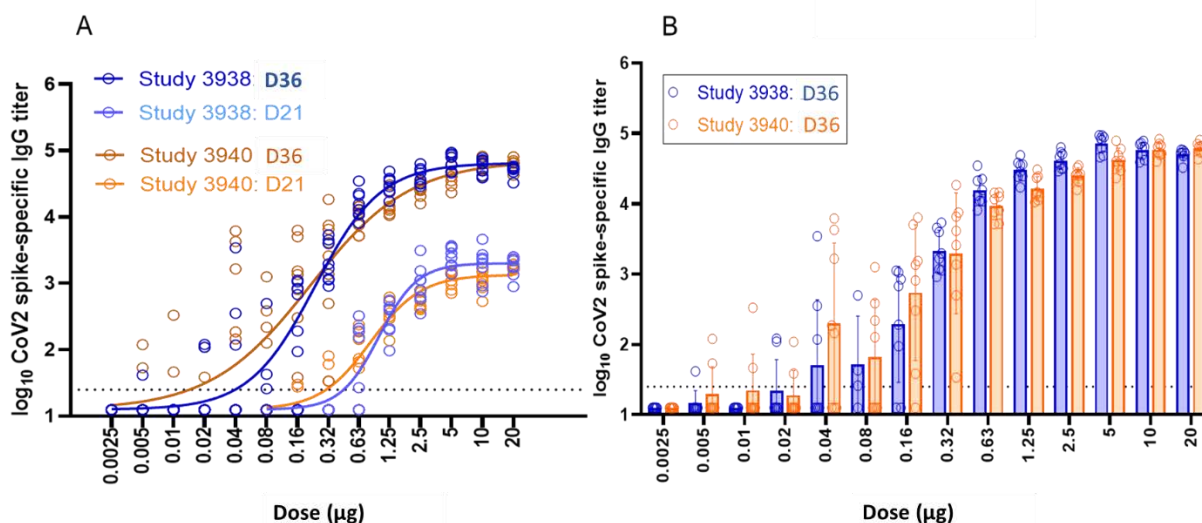
3. RESULTS

3.1. Evaluation of Humoral Immune Responses Across a Broad Range of mRNA-1273 Dose Levels (MOD-3938/3940)

SARS-CoV-2 S-specific IgG was measured by ELISA in Day 21 and Day 36 serum in Study MOD-3938 and Study MOD-3940.

BALB/c mice immunized with varying doses of mRNA-1273 exhibited consistency across the 2 independent studies, with similar IC_{50} (Figure 1A), and titer levels (Figure 1B). As expected, based on past studies, the range of titers within a group is smaller at higher dose levels and larger at lower dose levels.

Figure 1: SARS-CoV-2 Spike-Specific IgG Titer in Studies MOD-3938 and MOD-3940



Abbreviations: CoV2 = SARS-Cov-2; D = day; IgG = immunoglobulin G; SARS = severe acute respiratory syndrome; SARS-Cov-2 = 2019 novel coronavirus.

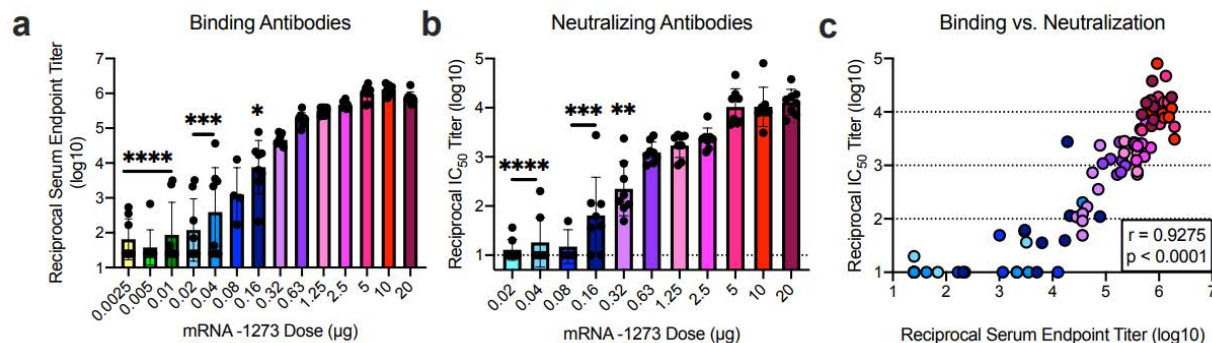
Note: Figure A shows 4-parameter curve of log₁₀ titer vs log₁₀ dose at both Day 21 and Day 36.

Figure B shows mean log₁₀ titer and standard deviation for each dose level at Day 36.

Geometric means or arithmetic means are represented by the heights of bars or symbols; error bars represent the corresponding SD.

In Study MOD-3938, neutralizing antibodies were measured in Day 36 serum against a homotypic SARS-CoV-2 pseudovirus. mRNA-1273 elicited a dose-dependent neutralizing antibody response (Figure 2B). Analysis suggests that there is a significant correlation between binding (Figure 2A) and neutralizing antibody titers in Day 36 serum (Figure 2C).

Figure 2: SARS-CoV-2 Pseudovirus Neutralization and Correlation to Binding Antibody (Study MOD-3938)



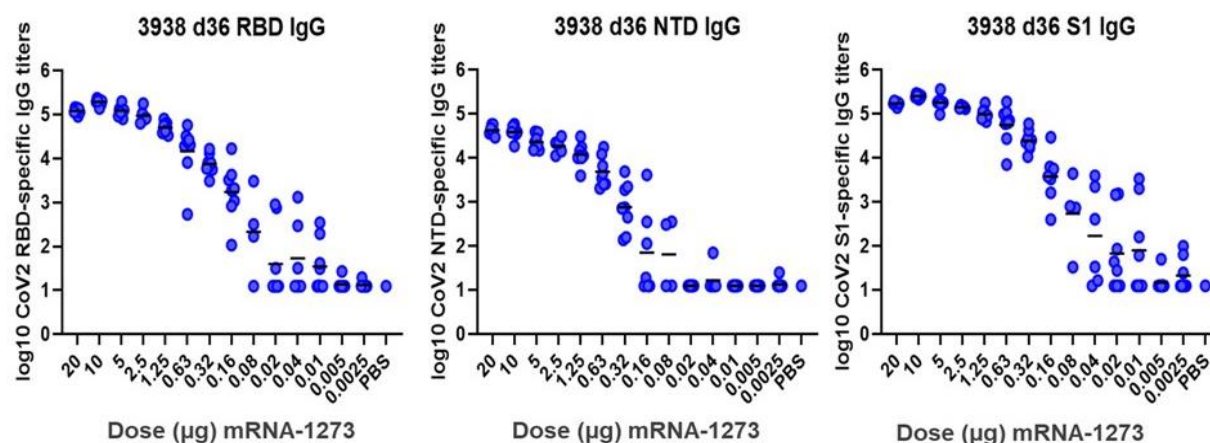
Abbreviations: IC₅₀ = half-maximal inhibitory concentration; SARS = severe acute respiratory syndrome; SARS-CoV-2 = 2019 novel coronavirus.

Note: Geometric means or arithmetic means are represented by the heights of bars or symbols; error bars represent the corresponding SD.

* p value < .05, ** p value < .01, *** p value < .001, **** p value < .0001.

Also in Study MOD-3938, binding IgG antibody titers were measured in the Week 5 (Day 36) serum against the RBD, NTD, and S1 subunits of the SARS-CoV-2 S protein. A dose-dependent response was observed for all antigens in vaccinated animals. binding IgG antibody titers against S1 and RBD IgG antibody titers were detected in some vaccinated animals, even at the low dose of 0.01 µg of mRNA-1273 (Figure 3).

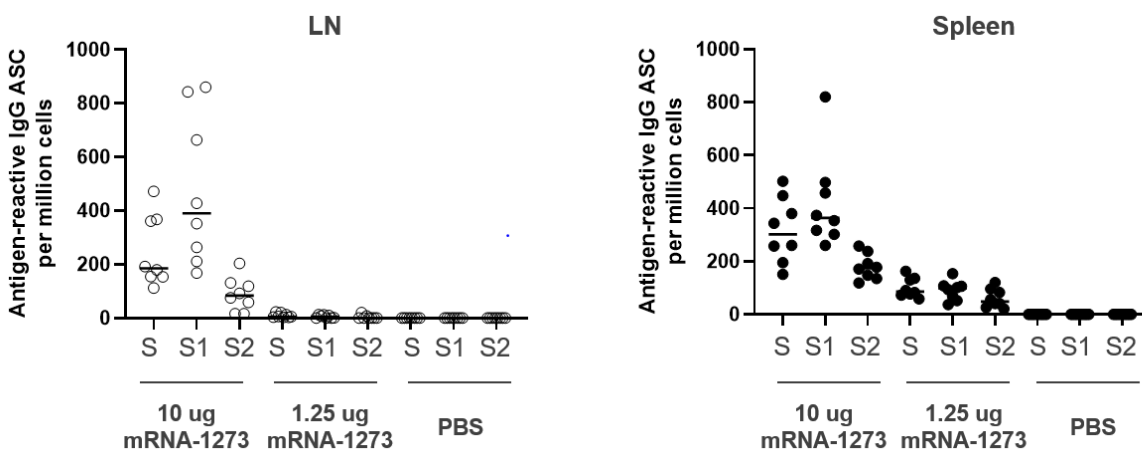
Figure 3: Day 36 SARS-CoV-2 Receptor Binding Domain, N-Terminal Domain, and S1 Subunit IgG ELISA Titers (Study MOD-3938)



Abbreviations: COV2 = SARS-CoV-2; d = day; ELISA = enzyme-linked immunosorbent assay; IgG = immunoglobulin G; NTD = N-terminal domain; RBD = receptor binding domain; S1 = S1 subunit of the SARS-CoV-2 protein; SARS = severe acute respiratory syndrome; SARS-CoV-2 = 2019 novel coronavirus.

Note: Bars are representative of the mean value.

In Study MOD-3940, antigen-reactive IgG ASCs were measured by ELISpot at Week 5 (Day 36) from the spleen and LNs of immunized mice. SARS-CoV-2 whole S protein as well as the S1 and S2 subunits were used as antigens. mRNA-1273 induced IgG antigen-reactive ASCs both in the spleen and LNs in a dose-dependent manner (10 µg of mRNA-1273 compared to 1.25 µg). Lower responses to the S2 subunit compared to S1 subunit of the SARS-CoV-2 S protein were observed (Figure 4).

Figure 4: Day 36 IgG ELISpots in Lymph Nodes and Spleen (Study MOD-3940)

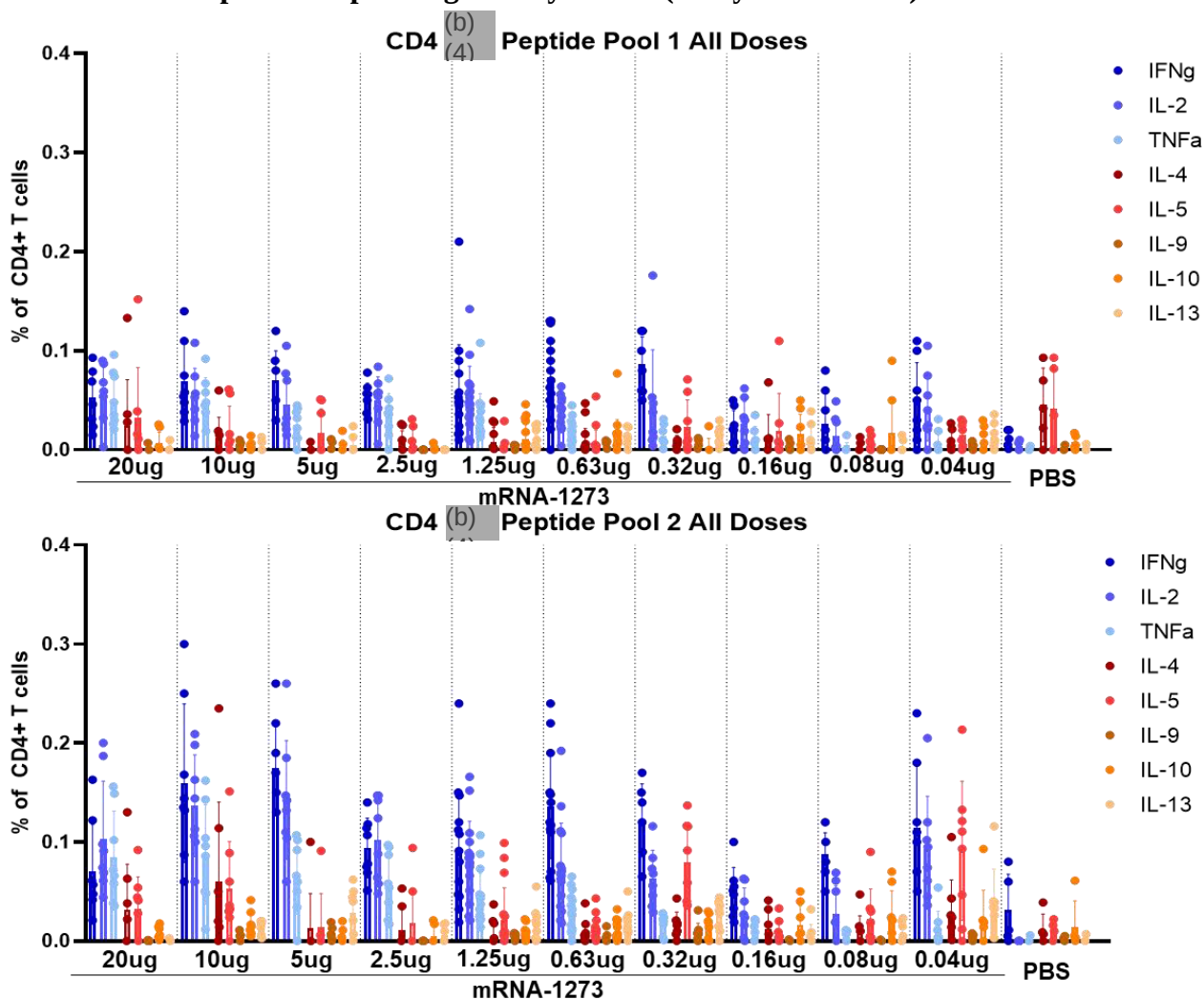
Abbreviations: ASC = antibody-secreting cell; ELISpot = enzyme-linked immunospot; IgG = immunoglobulin G; LN = lymph node; PBS = phosphate-buffered saline; S = spike; S1 = S1 subunit of the SARS-CoV-2 protein; S2 = S2 subunit of the SARS-CoV-2 protein.

Note: Bars are representative of the mean value.

3.2. Evaluation of Cellular Immune Responses Across a Broad Range of mRNA-1273 Dose Levels (MOD-3938)

Immunization with mRNA-1273 produced S protein-specific CD4⁺ (Figure 5) and CD8⁺ T-cell responses (Figure 6). CD4⁺ T-cell responses had a Th1-directed cytokine profile, as indicated by elevated levels of IFN- γ , IL-2, and TNF- α . A dose-dependent CD4⁺ Th1 cytokine response was not observed. Low CD4⁺ Th2 cytokine levels were detected in mice immunized with mRNA-1273 across all doses. Spike antigen-specific CD8⁺ responses decreased with decreasing dose levels.

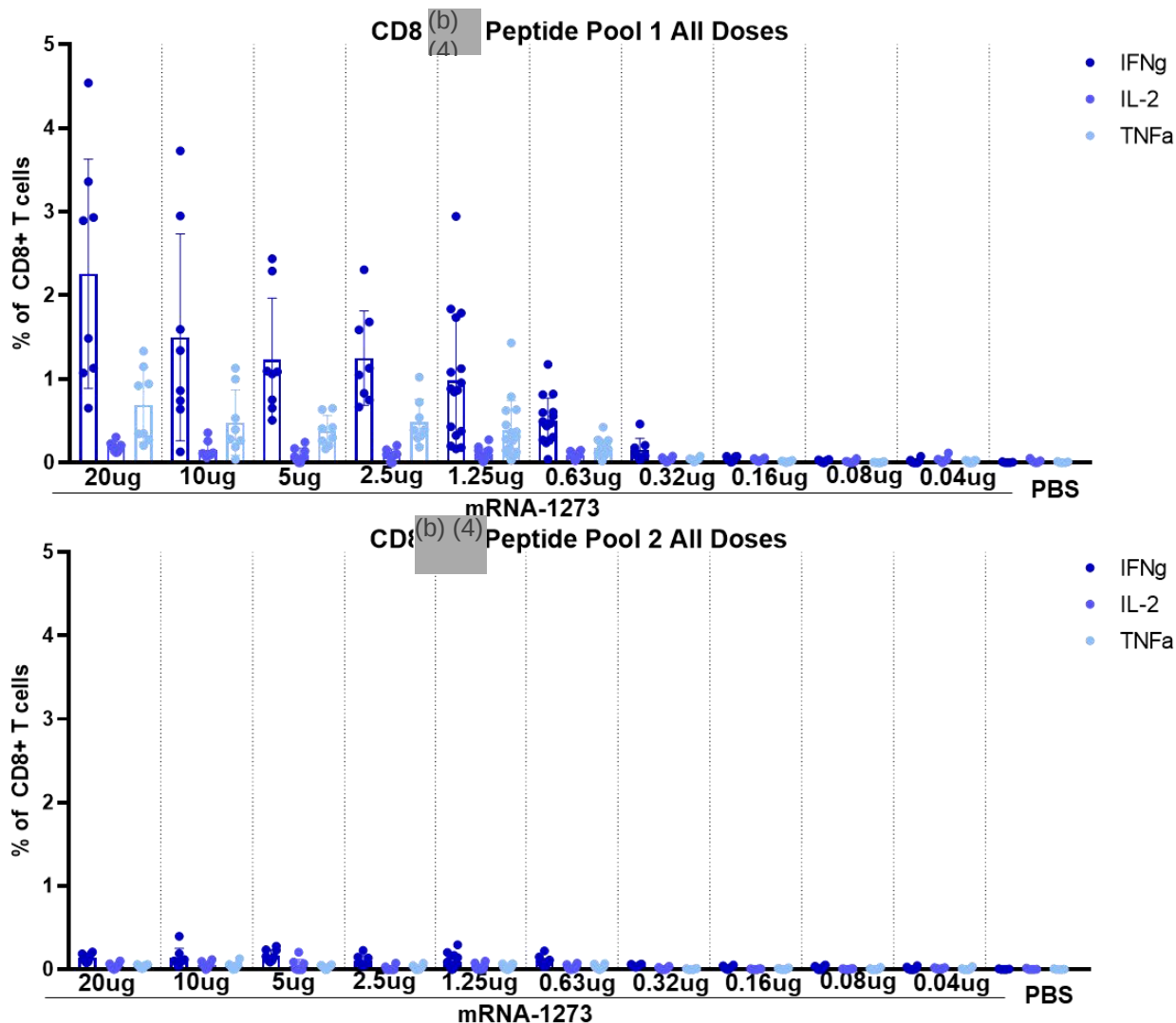
Figure 5: Immunization With mRNA-1273 Produces Spike-Specific CD4+ T-Cell Responses Expressing Th1 Cytokines (Study MOD-3938)



Abbreviations: IFN γ = interferon gamma; IL = interleukin; PBS = phosphate-buffered saline; Th-1 = T helper 1; TNF α = tumor necrosis factor alpha.

Note: Percentage means are represented by the heights of bars or symbols; error bars represent the corresponding SD.

Figure 6: Immunization With mRNA-1273 Produces Spike-Specific CD8+ T-Cell Responses Expressing Th1 Cytokines (Study MOD-3938)



Abbreviations: IFNγ = interferon gamma; IL = interleukin; PBS = phosphate-buffered saline; Th-1 = T helper 1; TNFα = tumor necrosis factor alpha.

Note: Percentage means are represented by the heights of bars or symbols; error bars represent the corresponding SD.

4. CONCLUSIONS

mRNA-1273 was immunogenic in BALB/c mice, showing a dose-dependent response in SARS-CoV-2 whole S protein-specific IgG binding antibody titers that was similar across the 2 studies (MOD-3938 and MOD-3940). In addition, RBD, NTD, and S1 subunit-specific IgG antibody titers as well as neutralization titers measured in Study MOD-3938 showed similar dose-dependent responses. There was a strong correlation between binding and neutralizing antibody responses. In Study MOD-3940, mRNA-1273 induced IgG antigen-reactive ASCs in both the spleen and LNs in a dose-dependent manner. In Study MOD-3938, in splenocytes stimulated with S1 and S2 subunit overlapping peptide pools encompassing the entire SARS-CoV-2 S protein, mRNA-1273 drove Th1-directed CD4⁺ T-cell responses and CD8⁺ T-cell responses. Although a dose-dependent CD4⁺ Th1 cytokine response was not observed, CD8⁺ T-cell response decreased with decreasing dose.

5. REFERENCES

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6. REPORT APPROVAL

Study Director:

See esignature and date signed on last
page of document

(b) (6)

ModernaTX, Inc.

Date

7. **APPENDICES**

[Appendix 1 Summary of Analysis: Lot AMPDP-200005 Release](#)

APPENDIX 1: SUMMARY OF ANALYSIS: LOT AMPDP-200005 RELEASE



200 Tech Square • Cambridge, MA 02139
phone 617-714-6500 • fax 617-583-1998

SUMMARY OF ANALYSIS

Product Description: mRNA-1273 Drug Product

Project Code: mRNA-1273 **Batch ID:** B-20200201-01404

Manufacturer: Moderna **Sample ID:** S-200201-00010

Moderna Lot Number: AMPDP-200005 **Date of Manufacture:** 01-Feb-2020

mRNA Lot Number: (b) (4) **Expiry Date:** N/A

Volume: 0.6 mL **Shipping Temperature:** -70°C

Storage Temperature: -70°C

Final Buffer 10.7 mM Sodium Acetate; 8.7% Sucrose; 20 mM Tris, pH 7.5

Container Closure: 2R vial, USP Type I; Traditional rubber stoppers are West 4432/50 (chlorobutyl rubber) composition with a Flurotec coating.

Test	Method	Target Attributes	Results
Identity	DPTM-0023	(b) (4)	
Appearance	DPTM-0030	White to off-white dispersion. May contain visible, white or translucent product-related particulates	Conforms, White to off-white dispersion; essentially free of visible particulates
RNA Content	DPTM-0024	(b) (4)	
RNA Purity	DPTM-0046	(b) (4)	
% Encapsulation	DPTM-0063	(b) (4)	
Filtered Mean Particle Size	DPTM-0012	(b) (4)	
Filtered Polydispersity Index		(b) (4)	



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Test	Method	Target Attributes	Results
Lipid Identification:			
SM-102	DPTM-0019	(b) (4)	
Cholesterol			
DSPC			
PEG2000-DMG			
Lipid Content (mg/mL)			
SM-102	DPTM-0019	(b) (4)	
Cholesterol			
DSPC			
PEG2000-DMG			
Lipid Impurities (% Area)			
Individual Impurities	DPTM-0019	(b) (4)	
Total Impurities			
pH	USP <791> DPTM-0028		
Osmolality	USP <785> DPTM-0029		
Bacterial Endotoxin	USP <85>		
Bioburden	USP<61>		
Particulate Matter	USP <788> Method 2		



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REVISION HISTORY

Revision #	Change Details	Author	Effective Date
1.0	Introduction of a New Document	(b) (6)	Date of Approval
2.0	Added in results for Identity and particulate matter	(b) (6)	Date of Approval
3.0	Added in results for bioburden testing and updated bacterial endotoxin results with values from ACCI (DPAD-00577)	(b) (6)	Date of Approval
4.0	Added correct test methods for RPIP and (b) (4)	(b) (6)	Date of Approval

Signature Page for VV-NC-000833 v1.0

Approval	(b) (6) 19-Oct-2020 14:22:57 GMT+0000
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